

http Hypertext transfer protocol. The call and response system used to link websites to the *Internet*.

Hubble's law This law states that a galaxy's distance is proportional to the speed at which it is moving away from us; the more distant the galaxy, the faster it is receding. This shows that the Universe is expanding.

Human Genome Project The worldwide science project completed in 2003 to map the entire sequence of genes in human DNA.

hydraulic pressure The pressure created by a fluid, for example, when pushed through a pipe.

hydraulics The study and phenomena of liquids flowing through pipes, especially when used as a source of power.

hydrocarbon A chemical compound made up of carbon and hydrogen only.

hydrogen The lightest, most abundant chemical, which makes up about 75 percent of the total mass of elements in the Universe.

hydrostatics The branch of physics that studies the pressure and equilibrium of liquids at rest.

imaginary number Any number that is a multiple of the square root of -1 , which does not exist as a "normal" number.

imaging Any method of producing images, especially when done indirectly by analyzing X-rays, magnetic responses of materials, etc.

immune system The body's natural defense mechanisms, which react to foreign material such as microorganisms, with effects such as inflammation and antibody production.

immunization The priming of the body's immune system, by *inoculation*, to fight against future infection.

indeterminate equation A mathematical equation which has more than one solution.

induction Any of the processes by which (1) an object becomes electrically charged when near another charged object; (2) a magnetizable object becomes magnetized when in the presence of an electric field, including one produced by an electric current; (3) an electric current is produced in a *circuit* by a varying magnetic field.

inertia The tendency of an object to remain at rest or to keep moving in a straight line until a force acts on it.

infrared radiation A type of electromagnetic radiation with a wavelength just longer than that of visible light but shorter than that of microwaves. It is often experienced as heat.

inheritance The pattern or manner in which genetic characteristics are passed on.

inhibitor In chemistry and biology, a substance that prevents or hinders a reaction or physiological response.

inoculation The deliberate introduction of disease-causing organisms into the body in a mild or harmless form to stimulate the production of antibodies that will provide future protection against the disease.

inorganic chemistry Branch of chemistry that deals with all chemicals except the large number of organic compounds (those that contain carbon-hydrogen bonds).

insulator A material that stops or reduces the flow of electricity, heat, or sound.

integrated circuit A tiny electric circuit made of components built into the surface of a silicon chip.

interference The disturbance of signals where two or more waves meet.

interferometry Techniques for analyzing the interference patterns of waves.

Internet The electronic information network linking computers around the world.

interstellar space The space between stars, where density of matter is typically very low.

ion An atom or molecule that has lost or gained one or more electrons to become electrically charged.

ionic bond A chemical bond formed when one or more electrons have been transferred from one atom to another, creating two ions of opposite charge that attract each other.

ionosphere The part of Earth's atmosphere that reflects radio waves. It lies within the thermosphere.

irrational number Any number that cannot be expressed as one whole number divided by another.

isomer A chemical compound with the same formula but a different structure to another compound.

isotope A version of a chemical element in which the atoms have different numbers of neutrons in their nuclei compared with other atoms of that element.

IVF In-vitro fertilization. The techniques (informally called the "test-tube baby" method) of arranging for a sperm to fertilize an egg outside the body before implanting the early embryo back in the womb.

joule The SI unit of work or energy.

karyotype Chromosomal characterization of a species or individual in terms of the number, size, and structure of each chromosome; also, a diagram that shows this.

Kelvin scale See *absolute scale*.

kinetic energy The energy an object has because of its movement.

Kyoto Protocol An international agreement on climate change that sets industrialized countries binding targets for reducing greenhouse gas emissions.

Lamarckism The theory that evolution depends on inheritance of characteristics acquired during an organism's life.

laser A device used to produce an intense narrow beam of light in which the light rays are parallel.

latent heat The heat absorbed or given out, without change of temperature, when a substance changes between a liquid and a gas, or between a solid and a liquid.

lathe A machine designed to spin objects around while cutting them into shape.

latitude A measure of distance from the equator (the poles are at 90° latitude and the equator is at 0°). Lines of latitude are imaginary lines drawn around Earth, parallel with the equator.

lens A transparent object shaped to refract light so that it produces or causes to be produced a sharp image.

lepton A family of fundamental particles, including the electron, that are not affected

by the strong nuclear force, unlike quarks and particles made up of quarks.

leucocyte A white blood cell.

Leyden jar *Capacitor*, invented in the 18th century, capable of delivering electric shocks.

line of force One of the imaginary lines in a field of force.

linear equation A mathematical equation that contains no variable number multiplied by itself (for example, there is no x^2 , x^3 , etc.). Linear equations result in straight lines when plotted out on graphs.

lithosphere The rigid outer layer of Earth, consisting of the crust and the uppermost layer of the underlying mantle.

lock and key Term applied to a situation (for example involving biological molecules) where two parts have to match and interact like a lock and key to effect a change.

lodestone A naturally magnetic piece of the iron-containing mineral magnetite.

logarithm In mathematics, the *power* to which a *base*, such as 10, must be raised to yield a given number.

long count An indefinitely long calendar with a starting point of several thousand years ago, used by the Maya and other Mesoamericans.

longitude A measure of position on Earth, measured by degrees east or west from an imaginary line (the prime meridian) running from the north to the south pole via Greenwich in London. All other lines of longitude also run from the north to the south pole.

longitudinal wave A wave whose to-and-fro movement takes place along the line that the wave is traveling, not at right angles to it. Sound waves are an example.

low frequency Involving a relatively small number of vibrations in a given time period.

luminosity The amount of light given out by an object, such as a star.

lunar eclipse See *eclipse*.

lymphatic system A network of tubes and small organs that drains a fluid called lymph from the body's tissues into the bloodstream.

lymphocytes Types of white blood cell that play specialist roles in the immune system.

magnetic dip The downward-pointing angle at which a freely moving *compass* points, representing the fact that Earth's magnetic poles lie below the surface.

magnetic poles (1) The two regions of a magnet where magnetic effect is strongest. (2) The two variable points on Earth where Earth's magnetic field is strongest and toward which a *compass* needle points.

magnetism The invisible force of attraction or repulsion produced by a magnetic field.

magnetosphere The magnetic field around a star or planet.

mass The amount of matter in an object.

matter Anything that has mass and occupies space.

megabyte See *byte*.

megalith A large stone, especially one deliberately set in position as a marker or monument in prehistoric times.

meiosis A specialized type of cell division (strictly speaking, of nuclear division) that

takes place in two stages, and in which haploid sex cells are produced.

Mercator's projection A way of representing the Earth's surface on a flat map so that longitude and latitude lines are at right angles to each other.

merozoite A stage in the life cycle of some microscopic parasites.

mesopause The boundary between the mesosphere and the thermosphere, about 50 miles (80 km) above Earth's surface.

mesosphere (1) The layer of the atmosphere above the *stratosphere*. (2) The layer of the Earth's mantle below the *asthenosphere*.

metabolism The sum total of all the chemical reactions taking place in a living organism.

metal A substance typically having a combination of properties including shiny appearance, ability to be bent into shape, and high conductivity to heat and electricity. Most chemical elements are metals, and there are also thousands of metallic *alloys*.

metaphase The stage of *mitosis* and *meiosis* preceding *anaphase*, during which the chromosomes are aligned along the middle of the cell.

meteoroid Any rocky body, smaller than an asteroid, moving freely through space in the Solar System. If one falls to Earth and is not completely burnt up, it is called a meteorite.

microorganism A tiny organism which can be seen only with the aid of a microscope.

microscope An instrument that produces magnified images of very small objects.

mid-ocean ridge A ridge down the middle of the ocean floor, created by volcanic material erupting from the gap between oceanic plates. See also *plate tectonics*.

mitosis The division of the nucleus during normal cell division, in which each "daughter" nucleus has the same number of chromosomes as the parent cell.

mode (1) A particular pattern of vibrations. (2) In statistics, the value that occurs most frequently in a set of data.

model organism A specimen studied by scientists with a view to developing knowledge that can be applied more generally for understanding other organisms.

modular arithmetic Sometimes called clock arithmetic, a method of counting where one starts again at the beginning after a set point is reached.

modulation Transmitting information by superimposing an extra pattern upon a radio wave (called a carrier wave) or other waves.

molecule Smallest free unit of an element or compound, made up of at least two atoms.

momentum A quantity equal to an object's mass multiplied by its velocity.

Monocotyledons (Monocots) A major subgroup of flowering plants (including grasses, orchids, spring bulbs, palms, etc.) originally identified because they have only one cotyledon (seed-leaf) in their seeds.

motor nerve A nerve that transmits impulses from the central nervous system to operate a muscle or control a gland.

MRI Magnetic resonance imaging. A noninvasive form of medical imaging.